AE4868 – Numerical Astrodynamics

Mission Design - Lambert Targeter

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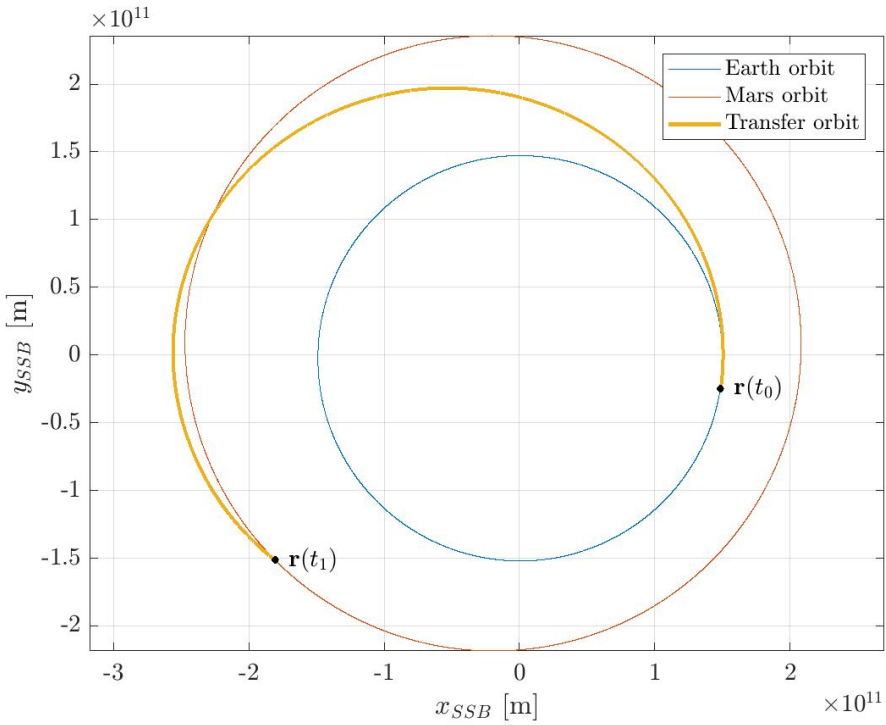
Lambert Targeter

Introduction

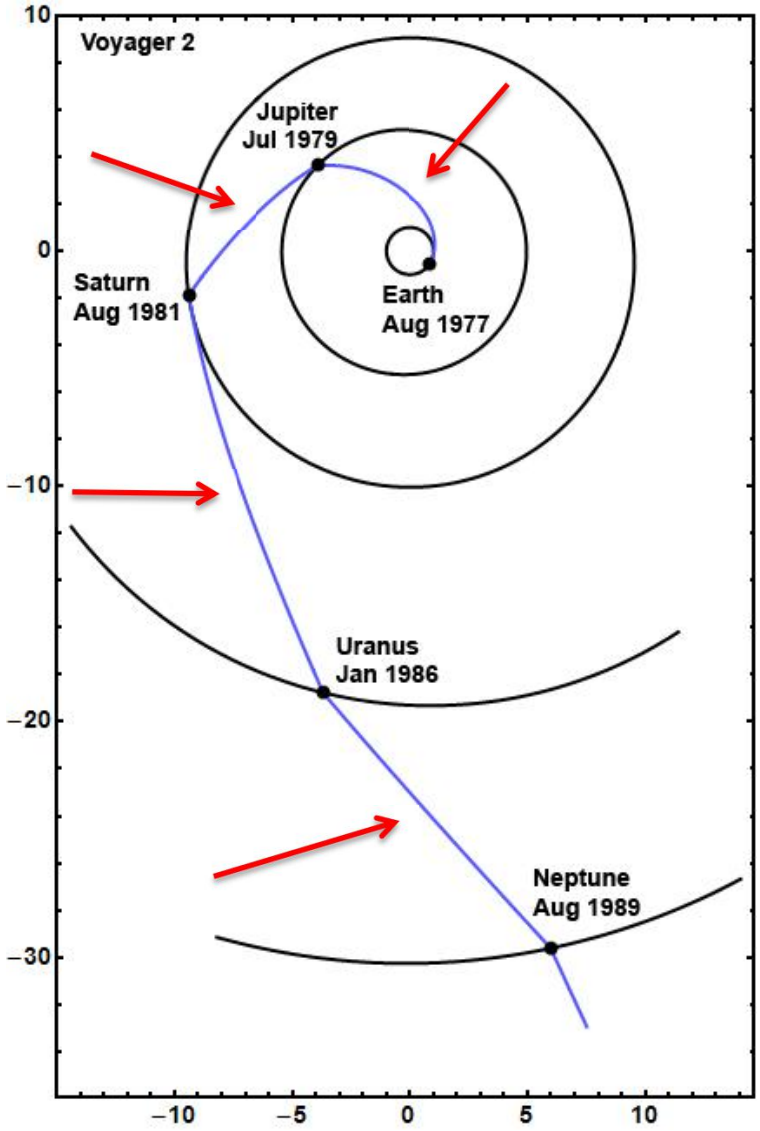
- **Lambert targeter**
  - A tool to solve **Lambert's problem** (see FoA)
  - A very quick, simple and (often) effective way to **start** high-thrust interplanetary mission design
- Typical **follow-up steps**
  - Use high-accuracy model
  - Lambert arc as initial guess



Lambert Targeter

Introduction

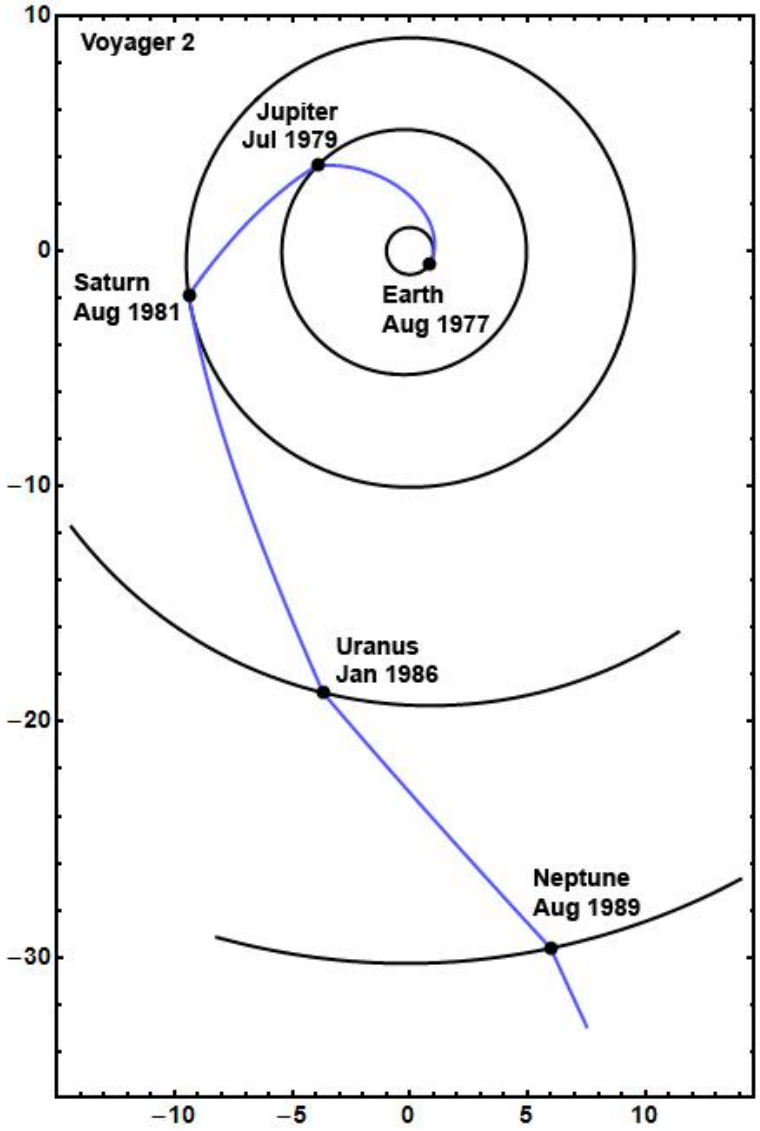
- This approach is typical in (preliminary) mission design
  - Start with, and optimize, a simple model
  - Use higher-fidelity model to verify feasibility of solution
  - Use higher-fidelity model to perform detailed design
- A Lambert targeter is a 'single leg' of a multiple gravity assist problem



Lambert Targeter

Introduction

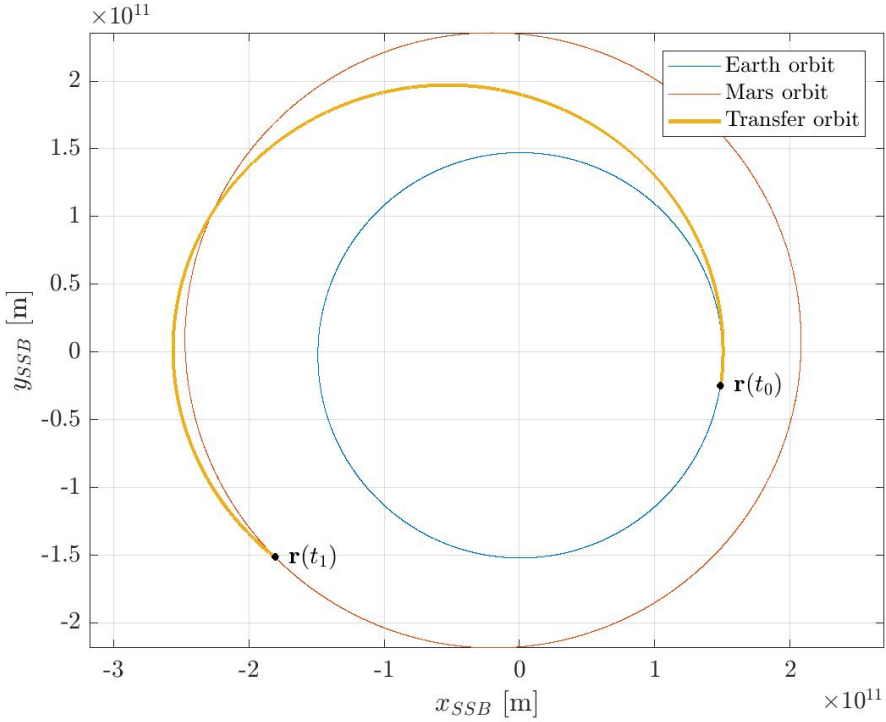
- ***This block of the course***
  - Use of analytical model as first guess for numerical design
  - Specific test case: Lambert targeter
- ***This series of lecture videos***
  - **Lambert targeter** - analytical and numerical considerations
  - **Variational equations** - definition and (quick) derivation
  - **Differential correction** - definition and application



Lambert Targeter

Introduction

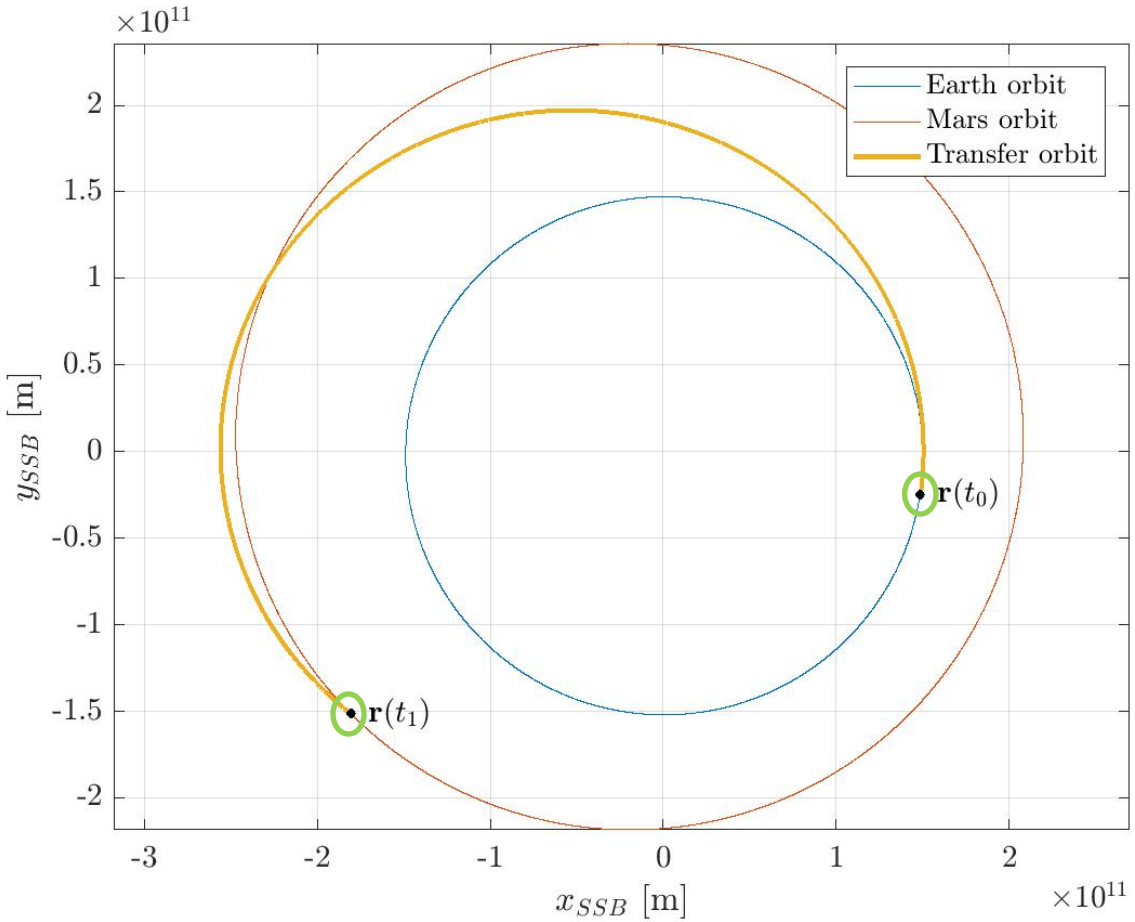
- ***This lecture***
  - **Lambert targeter** - basic assumptions and application
  - **Numerical propagation** - Lambert targeter as initial guess
  - **Numerical propagation** - influence of choice of initial propagation time



Lambert Targeter

Assumptions

- **Lambert targeter** - determine a Kepler orbit between two points



Lambert Targeter

Assumptions

- **Lambert targeter** - determine a Kepler orbit between two points

- Assumptions and input

  - Central body **located at the origin** (non-moving)
  - Only **central body point mass** exerts an acceleration
  - **Start position** w.r.t. central body is defined
  - **End position** w.r.t. central body is defined
  - **Time of flight** is defined
  - At most a **single revolution** around central body (in this case)
  - **Prograde** orbit

  (specific assumptions here)

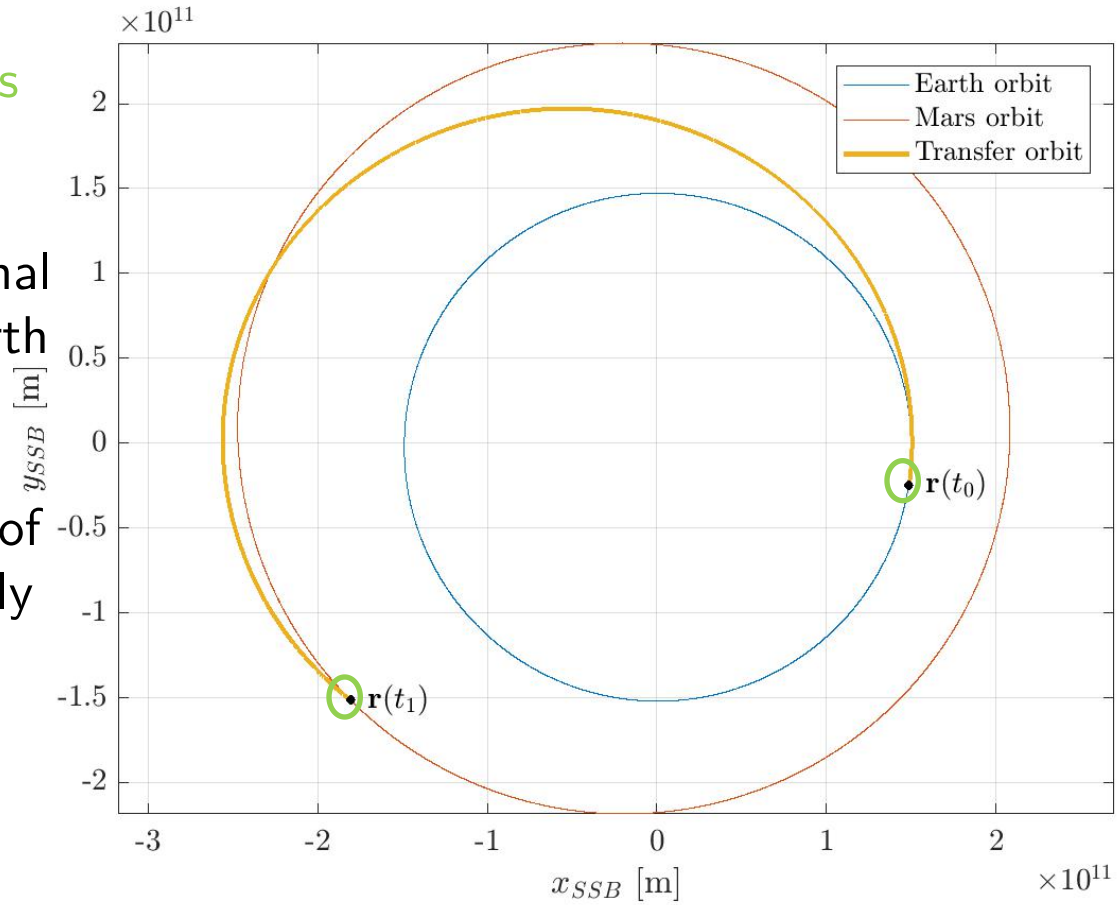
Lambert Targeter

Assumptions

- **Lambert targeter** - determine a Kepler orbit between two points

Example: Earth-Mars transfer; departure t_0; arrival t_1

- Earth and Mars states taken ephemerides
- **NOTE:** initial and final states are 'inside' Earth and Mars!
- Lambert arc (output of Lambert targeter) fully defined



Lambert Targeter

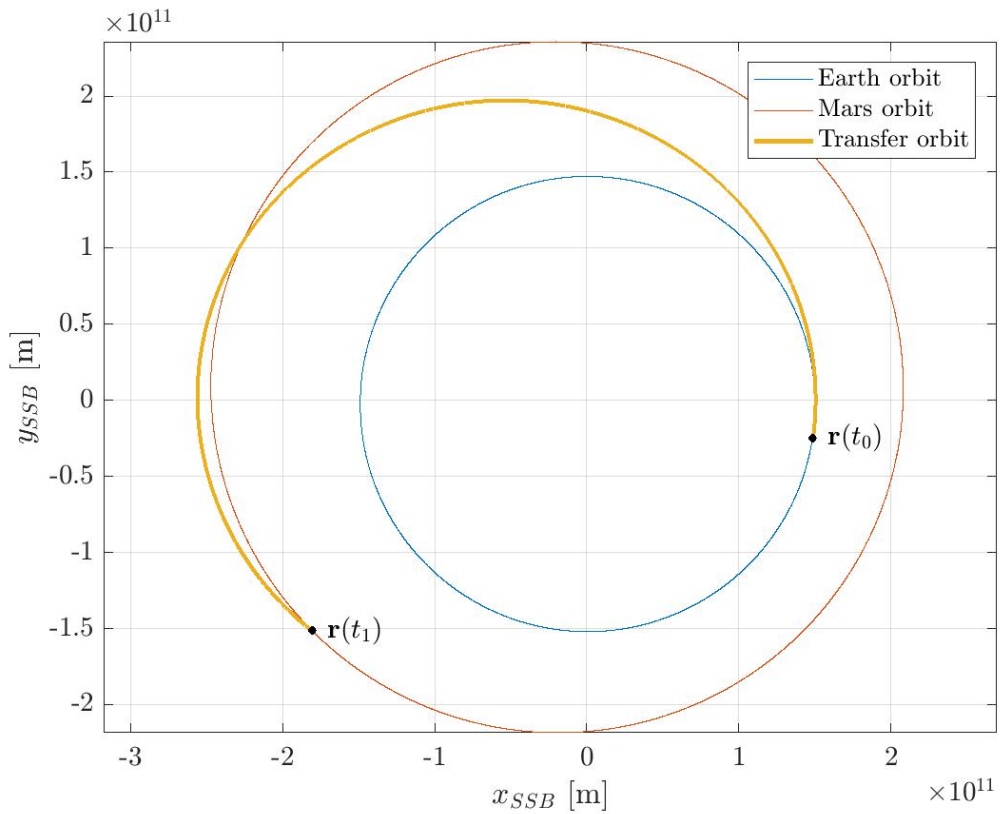
Assumptions

- Lambert targeter - to determine a Kepler orbit between two points
- **Our** assumptions and input define a unique solution (in most cases)
  - A **single** Kepler orbit exists between two points, for a given time of flight and number of revolutions
- Output
  - Kepler elements of transfer orbit
  - **Start** and **end velocity** w.r.t. central body (from Kepler elements)
  - Cartesian state along transfer (from Kepler elements)

Lambert Targeter

Link to numerical propagation - Problem setting

- Lambert targeter produces a Keplerian trajectory
  - **Reality:** The true trajectory will be perturbed
  - **Consequence:** We will never fly the exact Keplerian trajectory



Lambert Targeter

Link to numerical propagation - Problem setting

- Lambert targeter produces a Keplerian trajectory
  - **Reality:** The true trajectory will be perturbed
  - **Consequence:** We will never fly the exact Keplerian trajectory
- For interplanetary transfer, perturbations will typically be small (?)
  - Lambert targeter will be a 'reasonable' initial guess (?)
  - We can use Lambert arc as initial state for propagation
  - Difference between Keplerian and perturbed trajectory small (?)
- **What will be the issues when using Lambert arc as starting point?**

Lambert Targeter

Link to numerical propagation - Problem setting

- Our trajectory needs to meet boundary conditions
  - Position $\mathbf{r}(t_0)$ at initial time t_0
  - Position $\mathbf{r}(t_1)$ at final time t_1
- The Lambert targeter will meet both **by definition**

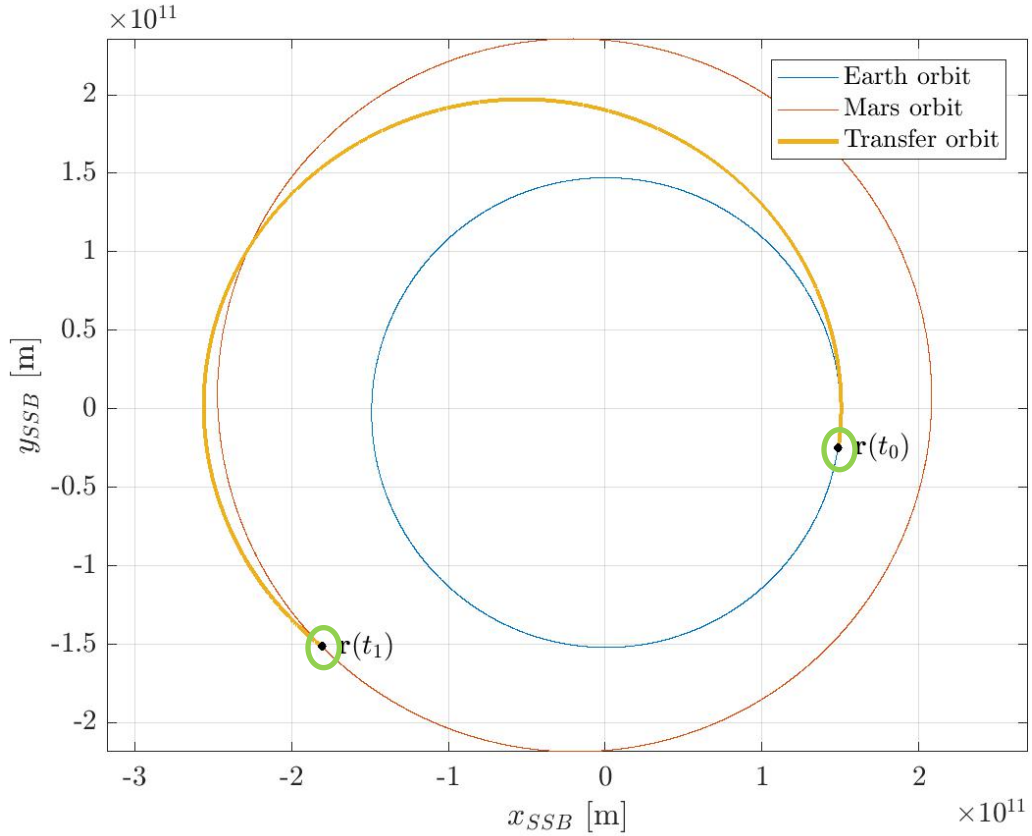
Lambert Targeter

Link to numerical propagation - Problem setting

- For Lambert targeter $\mathbf{r}(t_0)$ and $\mathbf{r}(t_1)$ are usually taken as the **centers of mass of our departure and arrival bodies**, respectively
  - We **cannot use these** for a numerical propagation!
- If we include Earth/Mars perturbations:

$$\mathbf{a} = -\frac{\mu}{r^2}\hat{\mathbf{r}}$$

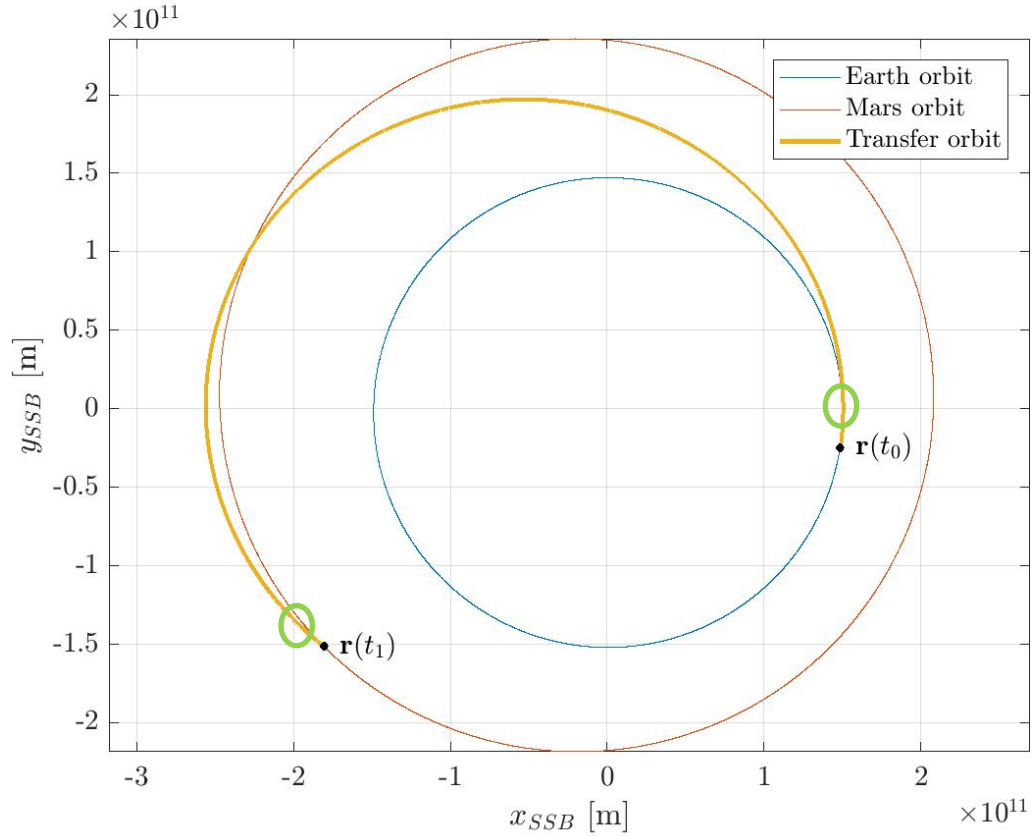
  - Distance $r \to 0$
  - Acceleration $\mathbf{a} \to \infty$



Lambert Targeter

Link to numerical propagation - Problem setting

- For Lambert targeter $\mathbf{r}(t_0)$ and $\mathbf{r}(t_1)$ are usually taken as the **centers of mass of our departure and arrival bodies**, respectively
  - Detailed design, assumption is modified - add 'buffer time' t
  - Required initial and final positions still taken from Lambert arc!



Lambert Targeter

Link to numerical propagation - Problem setting

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  - Detailed design, assumption is modified - add 'buffer time' t
  - Required initial and final positions still taken from Lambert arc!

- The role of these initial and final positions:
  - Where does 'real' trajectory have to start (Lambert arc at t_0+ t)
  - Where does 'real' trajectory have to arrive (Lambert arc at t_1- t)

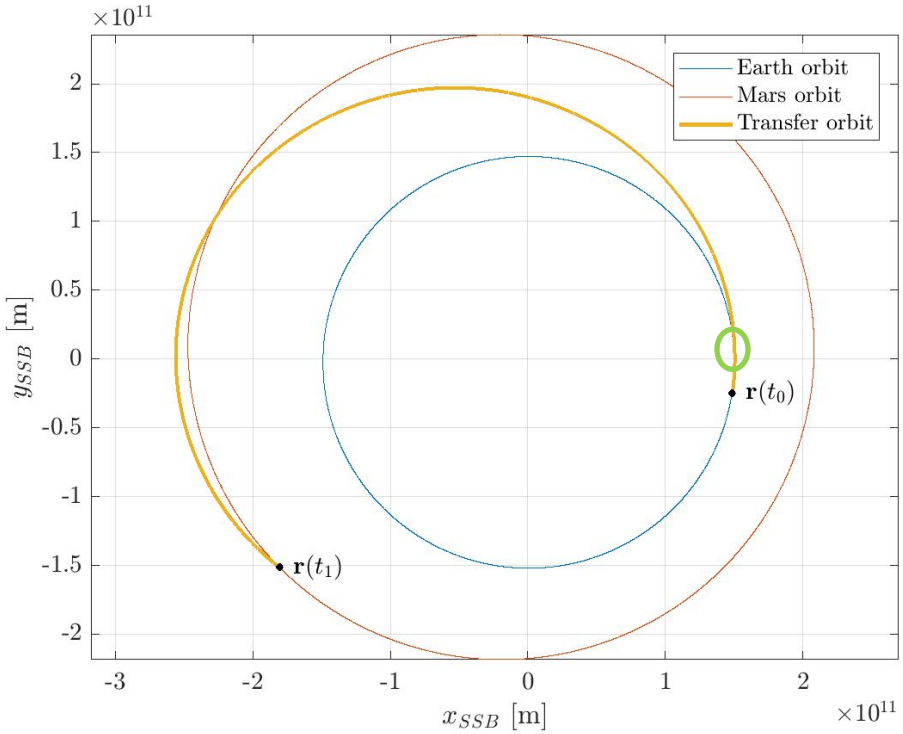
- By adding a buffer time, we are 'throwing away' the first and last bit of the Lambert arc
  - Departing 'close from' the Earth
  - Arriving 'close to' Mars
  - The departure and arrival phases are handled by separate analyses (not covered here)

Lambert Targeter

Link to numerical propagation - Problem setting

- Going from analytical model (Lambert arc) to a 'full' model (numerical propagation)

- How do we proceed?
  - There are various options, but let's look at some basic ones

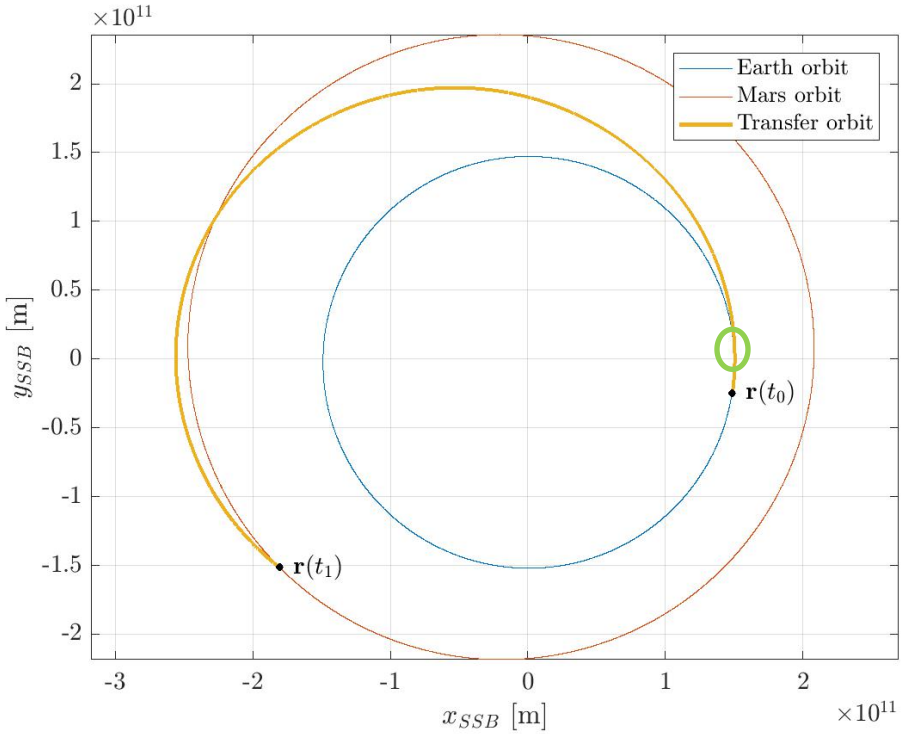


Lambert Targeter

Link to numerical propagation - Problem setting

- Going from analytical model (Lambert arc) to a 'full' model (numerical propagation)

- **Example: using Lambert targeter as initial state**
  - Extract Cartesian state from Lambert arc at $t_0 + t$
  - Use this as initial state $\mathbf{x}_0$ for numerical model



Lambert Targeter

Link to numerical propagation - Problem setting

- Going from analytical model (Lambert arc) to a 'full' model (numerical propagation)

- **Example: using Lambert targeter as initial state**
  - Extract Cartesian state from Lambert arc at $t_0 + t$
  - Use this as initial state $\mathbf{x}_0$ for numerical model
  - Define set of accelerations
  - Define integrator settings, dependent variables, *etc.*
  - Propagate dynamics

- **Output:**
  - Numerical state history $\mathbf{x}(\mathrm{t})$

- **Compare with**
  - Lambert arc state history $\bar{\mathbf{x}}(t)$

Lambert Targeter

Link to numerical propagation - Problem setting

- Going from analytical model (Lambert arc) to a 'full' model (numerical propagation)

- **Example: using Lambert targeter as initial state**
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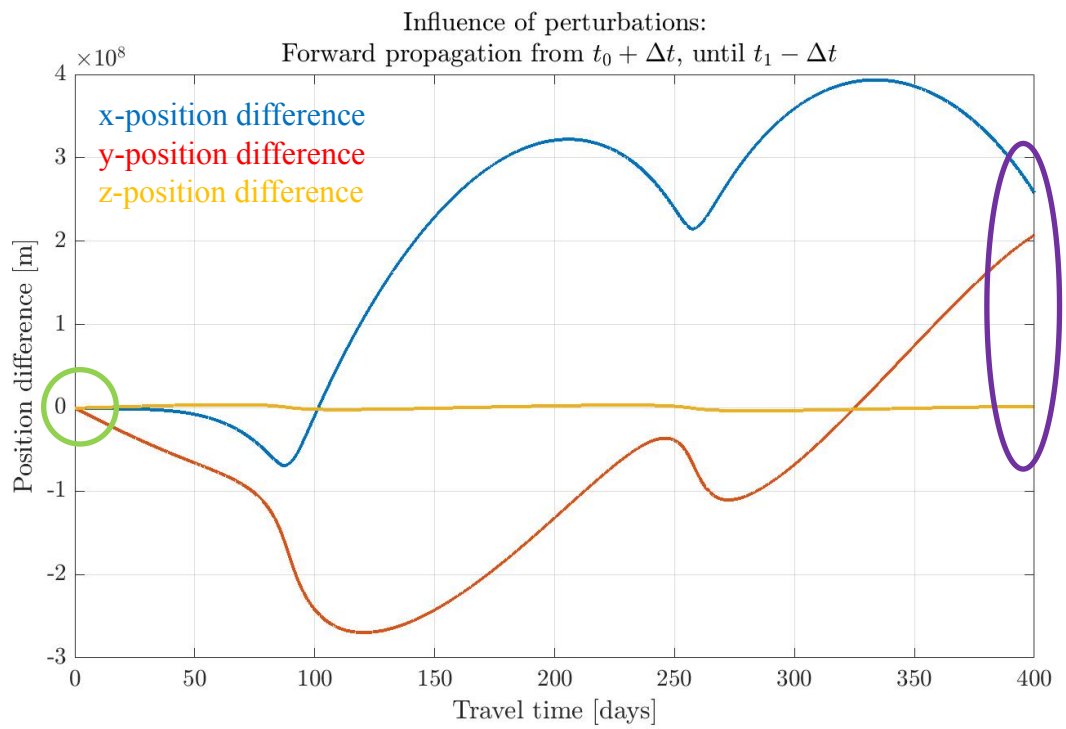
  - Use this as initial state $\mathbf{x}_0$ for numerical model
  - Define set of accelerations
  - Define integrator settings, dependent variables, *etc.*
  - Propagate dynamics

- **Analyze:**
  - How does the difference $\mathbf{x}(t) - \bar{\mathbf{x}}(t)$ behave?
  - This is a measure for the 'usefulness' of the Lambert arc in design

Lambert Targeter

Link to numerical propagation - results

- **Results:** Difference between analytical model (Lambert targeter) and numerical propagation
  - Initial position taken from Lambert targeter (at $t_0 + t$)
  - Difference between Lambert arc and numerical model will grow, but not always continuously!
  - Final numerical state is fairly close to Lambert arc



Lambert Targeter

Link to numerical propagation - results

- **Results:** Difference between analytical model (Lambert targeter) and numerical propagation
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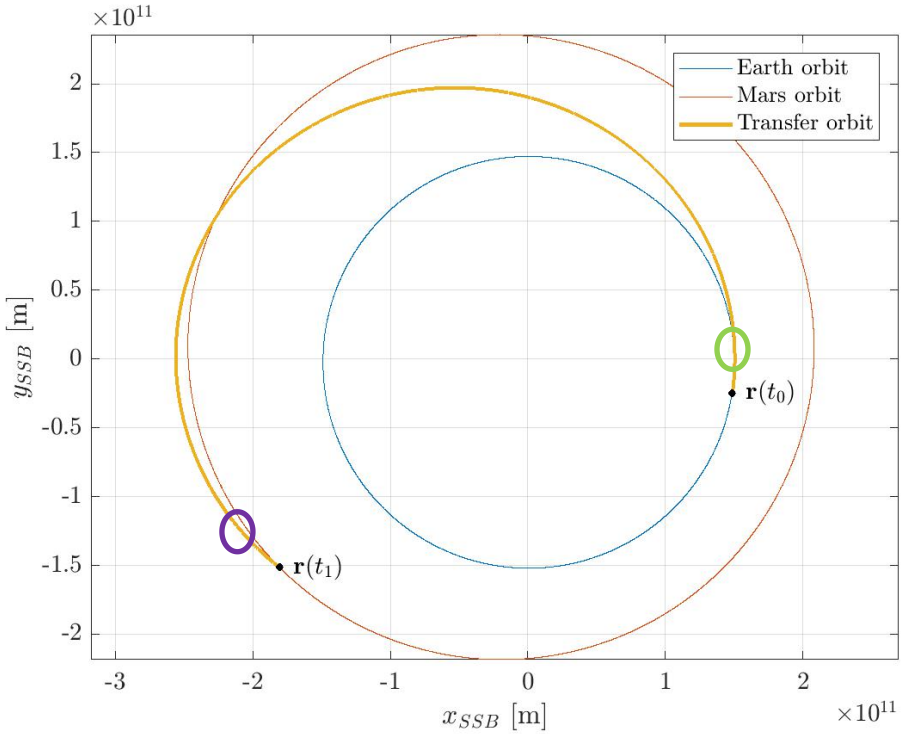
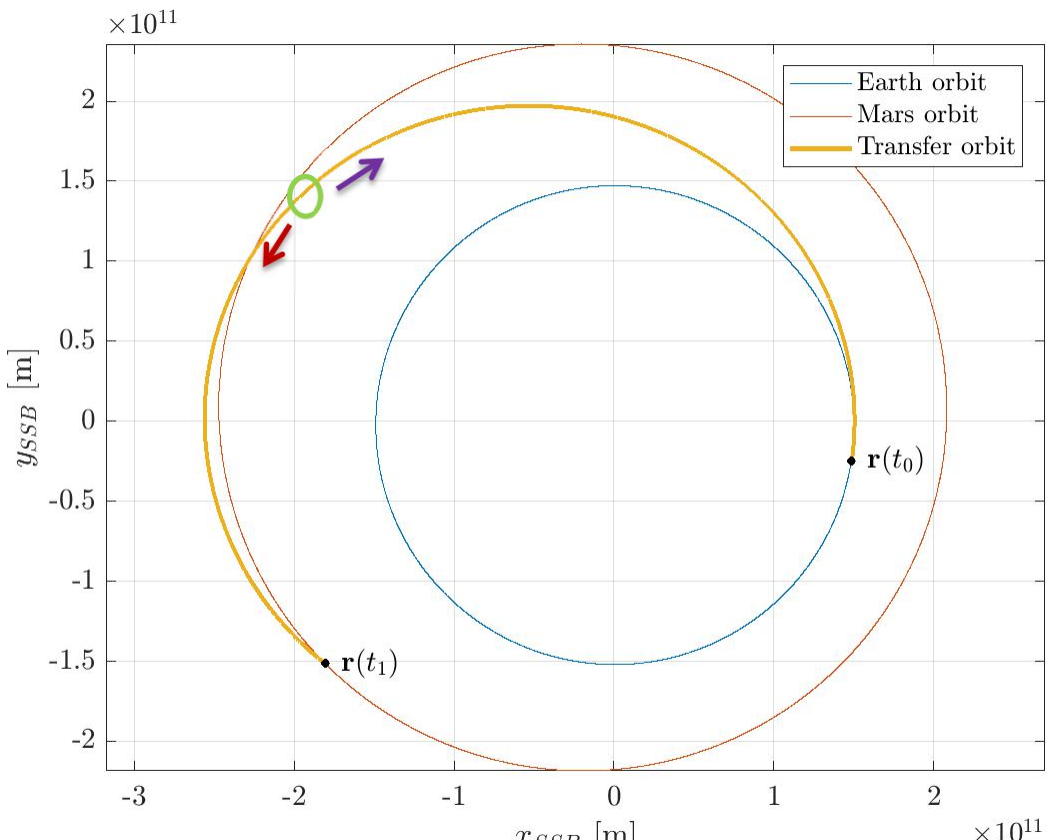


Lambert Targeter

Link to numerical propagation - choice of start time

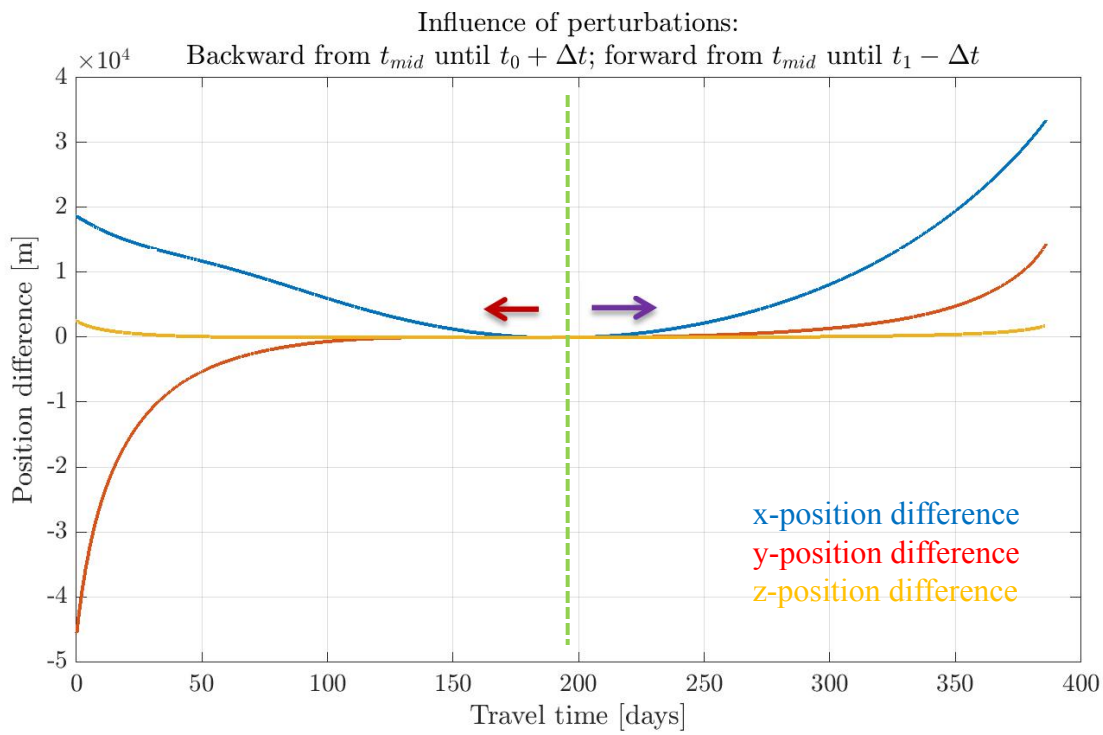
- We can choose any initial time T for numerical propagtion, and take $\mathbf{r}(T)$ from Lambert targeter as initial state
  - Instead of doing one propagation, we can do two!
  - One forwards, and one backwards



Lambert Targeter

Link to numerical propagation - choice of start time

- **Result**: propagating once forwards and once backwards from 'midpoint'
  - We will have deviations at start and end!
  - But, our deviations are smaller....



Lambert Targeter

Link to numerical propagation - summary

- Our trajectory needs to meet boundary conditions
  - Position $\mathbf{r}(t_0)$ at initial time t_0 (or with a buffer time t applied)
  - Position $\mathbf{r}(t_1)$ at final time t_1 (or with a buffer time t applied)
  - The Lambert targeter will meet both **by definition**
- We can choose any initial time T for numerical propagtion, and take $\mathbf{r}(T)$ from Lambert targeter as initial state
  - If we take $\mathbf{r}(t_0)$ from Lambert, we will not meet $\mathbf{r}(t_1)$
  - If we take $\mathbf{r}(t_1)$ from Lambert, we will not meet $\mathbf{r}(t_0)$
  - If we take $\mathbf{r}(T)$, with T between t_0 and t_1, we will meet neither $\mathbf{r}(t_1)$ not $\mathbf{r}(t_0)$

Lambert Targeter

Link to numerical propagation - summary

- We end up with two models

- **Analytical model - Lambert targeter**
  - Meets boundary conditions
  - Is not a realistic representation of reality

- **Numerical model - Tudat propagation**
  - Does not meet boundary conditions (but close!)
  - Is a realistic representation of reality

- **What we need**
  - Realistic representation of reality that meets boundary conditions
  - See coming videos!